

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
A-LEVEL · MATHEMATICS

MATHEMATICS 9164/2

PAPER 2 November 2022

3 hours

PAPER 2 — STRUCTURED QUESTIONS — CALCULATOR ALLOWED

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2022 9164/2.

Additional materials: A calculator may be used unless a question forbids it.

INSTRUCTIONS: Answer all questions. Full marks are not given for answers only — show method.

1. Find the stationary points of $y = x^3 - 6x^2 + 9x + 1$ and determine their nature. **[10]**
2. Find $\int x e^x dx$ using integration by parts. **[10]**
3. Solve $dy/dx = 3y$ given that $y = 2$ when $x = 0$. **[10]**
4. $x = 2t$, $y = t^2$. Find dy/dx in terms of t , and the equation of the tangent at $t = 1$. **[10]**
5. $A(1,0,2)$, $B(3,4,2)$. Find AB and $|AB|$. Show AB is perpendicular to $(2, -1, 0)$ or not. **[10]**
6. Show that $f(x) = x^3 - x - 1$ has a root in $[1, 2]$. Take one Newton-Raphson step from $x_0 = 1$. **[8]**
7. Solve $|2x - 3| < 5$. **[8]**
8. Solve $2 \sin \theta = 1$ for $0 \leq \theta \leq 360^\circ$. **[8]**
9. A GP has first term 5 and common ratio $1/2$. Find S_∞ and the smallest n with $S_n > 9.5$. **[8]**
10. $x^2 + y^2 = 25$. Find dy/dx at $(3, 4)$. **[8]**
11. Find the area between $y = 4x - x^2$ and the x -axis from $x=0$ to $x=4$. **[8]**

- 12.** For small θ in radians, write approximations for $\sin \theta$, $\cos \theta$, $\tan \theta$. Estimate $\sin 0.1$. **[8]**
- 13.** A particle starts from rest and accelerates at 2 m/s^2 for 6 s. Find final speed and distance. **[10]**
- 14.** Mass 3 kg at 4 m/s collides and sticks to 1 kg at rest. Find common speed. **[8]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).